JulleyOnline Collaborative Kanban — WebSocket (Socket.IO) Documentation

Connection URL: `http://localhost:5000`
Transports: `websocket`, `polling`
Authentication: JWT token passed in `socket.handshake.auth.token` or `Authorization` header

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1. Authentication & Tenant Room Authorization Flow

1. **Client Connection:**
2. The React frontend establishes a connection providing the user's JWT access token.
3. Socket.IO middleware verifies the JWT against `JWT_SECRET` and fetches the user's memberships.
4. If invalid, the connection is rejected with `Authentication error`.
5. **Joining Tenant-Scoped Rooms (`tenant:<tenant_id>`):**
6. Client emits `tenant:join` with `{ tenantId: "uuid" }`.
7. The server verifies that the user belongs to that specific tenant or is an `Org Admin`.
8. If authorized, the socket joins `tenant:<tenant_id>`.
9. A Hyderabad user update is **never** broadcast to Bangalore or Chennai rooms.

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2. Real-Time Events Catalog

Client → Server Events

Event Name	Payload	Description
`tenant:join`	`{ "tenantId": "uuid" }`	Requests to join an authorized tenant workspace room. Updates presence.
`disconnect`	*None*	Emitted automatically by Socket.IO on tab close or network drop. Cleans up presence.

Server → Client Events

Event Name	Payload	Description
`presence:update`	`{ "onlineUsers": [ { "id": "...", "name": "Rahul", "email": "..." } ], "count": 1 }`	Broadcast to `tenant:<id>` whenever a user connects, disconnects, or switches rooms. Multi-tab aware.
`task:created`	`{ "id": "uuid", "title": "...", "column_id": "...", "version": 1, ... }`	Broadcast to `tenant:<id>` when a teammate creates a new task.
`task:updated`	`{ "id": "uuid", "title": "...", "version": 2, ... }`	Broadcast to `tenant:<id>` when task title, description, or assignee is updated.
`task:moved`	`{ "id": "uuid", "column_id": "...", "position": 2, "version": 2, ... }`	Broadcast to `tenant:<id>` when a task is dragged to a new column or reordered.
`task:deleted`	`{ "id": "uuid" }`	Broadcast to `tenant:<id>` when a task is deleted.
`board:updated`	`{ "action": "created" | "updated" | "deleted", "board": {...} }`	Broadcast when a board or column structure changes.

3. Network Recovery & State Reconciliation

When a user loses network connectivity: 1. Socket.IO enters reconnection loop (`reconnectionAttempts: 10` , `delay: 1000ms`). 2. The UI shows a reconnecting indicator. 3. On reconnection, the client automatically re-authenticates and emits `tenant:join` . 4. The client immediately fetches authoritative board data from `GET /api/boards/:id` and `GET /api/tasks?board_id=:id` to reconcile any changes that occurred while offline.